

Pascal Pioneer

Dr Niklaus Wirth is considered the creator of the Pascal family of programming languages. Wirth earned a degree in Electronics Engineering from the Swiss Federal Institute of Technology, Zurich



Dr Niklaus Wirth (ETHZ), in 1959 followed by an M.Sc. from Laval University, Canada. He completed

his Ph.D. from the University of California, Berkeley, in 1963.

In 1968, Wirth returned to ETHZ as a professor of computer science. At ETHZ, Wirth developed Pascal (1970), Modula-2 (1979) and Oberon (1988). Pascal turned out to be a new computer language as well as a novel programming paradigm. It became an educational tool that motivated systematic and structured ways of thinking.

From 1992 onwards, his research has concentrated on digital design tools as well as the use of programmable devices.

Wirth has received several honorary doctorates and awards, including the Turing Prize, the IEEE Computer Society's Computer Pioneer Award, and the ACM SIGSOFT Outstanding Research Award in Software Engineering.

Wirth retired from ETHZ in 1999, after more than 30 years of service as a university professor. He has published several books that are considered classics in the field.

STOP THE GYPSY MUSIC

A Very Selective Virus

A new computer virus reported in Romania deletes only electronic gypsy music files. The Win.32. Antiman.A virus destroys audio and video files containing 'manele' music. Manele is traditional gypsy music played in an electronic style, which has become very popular in Romania. But it also has many detractors who have praised the virus as a utilitarian blessing.

Softwin, a Romanian company specialising in anti-virus software, said the worm works by targeting files associated with singers' names. Softwin spokesman Cosmin Mares said, "The effects of the virus are an increased usage of the Internet bandwidth and serious problems to companies' servers. All these cost money and someone has to pay."

Vali The Blizzard, a popular Manele singer, said, "I don't believe this is a threat for us. On the contrary they are doing us a favour. This way our fans will have to buy the CDs and not take the music from the Internet for free."



PIXEL EYES

Artificial Retina To Help Blind See Again

Scientists are helping the blind see again, one pixel at a time. If all goes well, an artificial retina could be commercially available within three years.

Artificial retinas have been successfully implanted in six patients, allowing them to see light and detect motion, researchers of the Association for Research in Vision and Ophthalmology in Florida announced in May.

Developed by researchers from the University of Southern California and the Doheny Eye Institute, the artificial retina pairs a tiny electronic eye implant with a video camera mounted on a pair of sunglasses.

The implant, a four-by-four grid of electrodes, connects to damaged photoreceptors on the patient's retina. The electrodes stimulate the photoreceptors, which transmit signals to the brain through the optic nerve.

The videocam translates the field of view into electrical impulses that are transmitted wirelessly to a microchip located behind the ear. In turn, the microchip is connected to the retinal implant by wires under the skin.

The system, named 'Argus' after the Greek god who had a hundred eyes, works only with patients with degenerated rods and cones, a condition often